Conductors and Insulators:
[or, non conductors or di-electric]

(1) **Conductors** are substances which have free charge ~~carrier~~ carriers. In metallic conductors, free charge carriers are free electrons. Valence electrons in metal atoms get detached from their parent atoms & move randomly inside metal.

When a potential difference or electric field ($\vec{E}$) is applied, the free electrons ($-e$) experience electric force ($\vec{F_{el}} = -e\vec{E}$) opposite to applied electric field & move opposite to electric field. This drift (unidirectional motion) of free electrons inside a conductor under action of external electric field constitute electric current in metallic conductors.

(2) **Insulator or Dielectrics** are substances which have no free electrons or no free charge carriers. So no electric current is observed in them even if external electric field or potential difference is applied.

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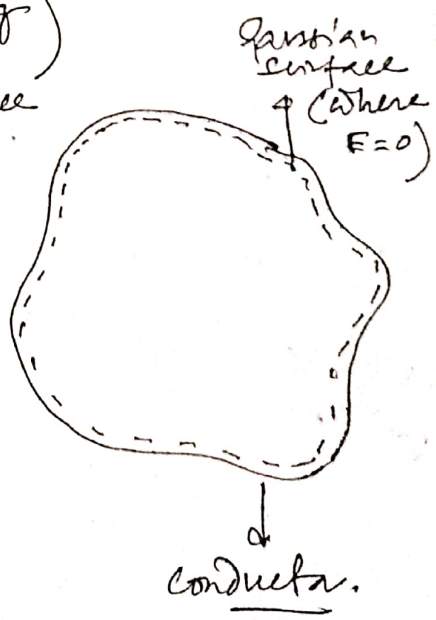
Properties of Conductor. (Imp for JEE)

① Electric field inside a conductor (neutral or charged) under electrostatic condition is zero. $\boxed{E_{inside\ conductor} = 0}$

explanation: ~~So~~ Since no electric current is found to exist inside an isolated (not connected to a potential difference like a battery) conductor, so we can say that no electric force ($F_{el} = qE$) is exerted on free charge carriers inside it. ($\because F_{el} = qE = 0 \therefore E = 0$) That means electric field E inside a neutral / charged isolated conductor is zero.

② Board Exam: electric charge always reside on the outer surface of a conductor.

Method-1 Let us ~~con~~ consider a Gaussian surface just inside the surface of a conductor (as in fig)



Since this Gaussian surface ~~is~~ (dotted line) is inside the conductor, so electric field on it is $E = 0$

By Gauss law on this dotted surface,

$$\oint \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\epsilon_0}$$

$\because \vec{E}_{inside\ conductor} = \vec{0}$

$\therefore q_{in} = 0.$

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ie no charge is present inside this gaussian surface.
Now as this gaussian surface is just inside & infinitely close ~~too~~ to the outer surface of the conductor, so any charge on conductor must be on the outer surface of conductor.

(3) JEE & Board | Surface of a charged conductor is an equipotential surface |

explanation: If there would be a potential difference between any two points of on the surface of a charged conductor, charge on its surface, would move from one point to other.

But as the charges on the outer surface is at rest, So there is no potential difference between any two pts on the surface of conductor. That means all pts on the surface of a conductor have same potential.

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(4) We know that electric field is perpendicular to equipotential surface. Since surface of a charged conductor is equipotential, so electric field and hence electric line of force is perpendicular to surface of a charged conductor.

JEE board

[see note on equipotential surface]

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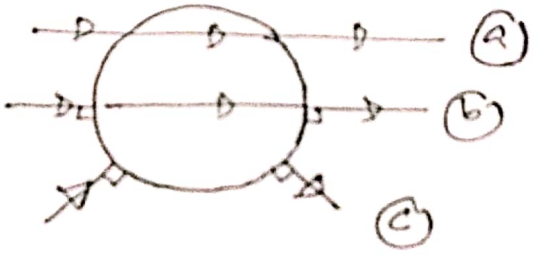
(5) Since electric field inside a neutral / charged conductor is zero, So no electric field line exists inside a conductor.

Application of (4) & (5)

(6) ex:



[Conductor in electric field.]

which field line is correct among (a) (b) (c)

Answer (c)

[Reason electric field inside a conductor = 0, & $\vec{E}$ is ⊥ to surface of conductor]

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V.V.Imp.

Inner side of a charged conductor.

a) is an equipotential region and

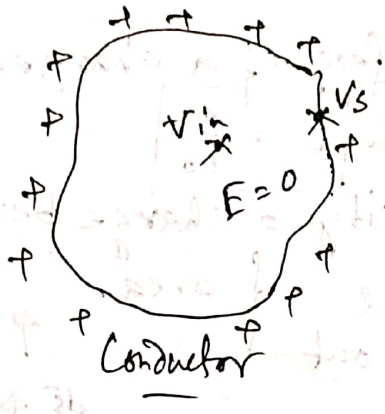
b) Potential at any point inside is equal to Potential on its surface.

Reason: Since electric field 'E' inside a charged conductor is zero

a) So potential inside is constant

$$\left[-\frac{dV}{dr} = E \quad \because E = 0 \quad \therefore V = \text{constant} \right]$$

b)



Let V_{in} & V_s be the potential inside & potential on surface.

Since $V_{in} - V_s = -\int \vec{E} \cdot d\vec{r}$

& $\vec{E}_{inside} = \vec{0}$

$\therefore V_{in} - V_s = 0$

or $\boxed{V_{in} = V_s}$

That means If potential on surface of a charged conductor be 10V, then at all points inside in the potential is 10 V. & potential inside is constant at 10V So $E_{inside} = 0$.

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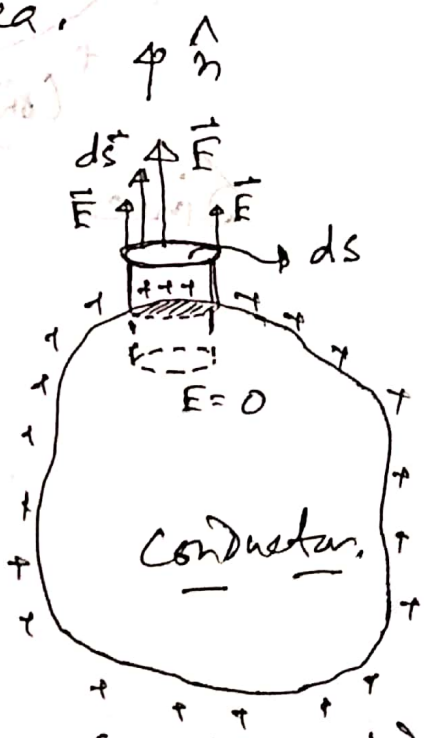
Imp Electric field at the surface of a charged conductor $\vec{E} = \frac{\sigma}{\epsilon_0}\hat{n}$ & $\boxed{E = \frac{\sigma}{\epsilon_0}}$ | Board Exam

σ = surface charge density = charge per unit area.

$\hat{n}$ = unit vector normal to surface in the outward direction.



we consider a pill box (a short cylinder) as the gaussian surface. Its crosssection is very small 'ds' & height is negligibly small. It is partly inside & partly outside.

∵ Inside the conductor electric field E = 0. & just outside electric is normal to surface with magnitude 'E'. (?)

Electric flux through the portion of this Gaussian cylinder inside the conductor is zero. (∵ E inside = 0)

over the curved surface of the Gaussian cylinder outside the conductor electric field E is tangential to curved surface. Hence no electric flux through the curved surface outside. But over small circular surface 'ds' of the Gaussian cylinder, situated outside the conductor, E is constant over ds & $\vec{E}$ is perpendicular to $d\vec{s}$ so flux through it is $d\phi = \vec{E}\cdot d\vec{s} = E\,ds\cos 0^\circ = E\,ds$

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The charge enclosed by this Gaussian pill box is $q_{inside} = \sigma ds$.

So By Gauss's Law

$$E ds = \frac{q_{in}}{\epsilon_0}$$

$$E \cancel{ds} = \frac{\sigma \cancel{ds}}{\epsilon_0}$$

$$\boxed{E = \frac{\sigma}{\epsilon_0}}$$

& $\vec{E} = \frac{\sigma}{\epsilon_0} \hat{n}$.

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